INDICES

Do the problem without using a calculator. Whenever appropriate, check answers by using a calculator

1. Evaluate the following:
   (a) 4^{-2}
   (b) $49^{\frac{1}{2}}$
   (c) 9^{-1}
   (d) $25^{\frac{3}{2}}$
   (e) $\left(\frac{1}{5}\right)^{-1}$
   (f) $\left(\frac{1}{7}\right)^{0}$
   (g) $\left(\frac{49}{81}\right)^{\frac{1}{2}}$
   (h) $\left(\frac{125}{8}\right)^{-\frac{1}{3}}$

2. Write each express below as a power of 2:
   (a) $8^3 \times 16^2$
   (b) $\frac{16\times 32^2}{8^3}$
   (c) $16\sqrt{2}$

3. Simplify the following
   (a) $(2a^2)^3$
   (b) $(5ab^2)^2$
   (c) $\sqrt{(64a^4}$
   (d) $\sqrt{(25a^2b^6)}$
   (e) $(2a^3)^5$
   (f) $(3m^2n^4)^4$

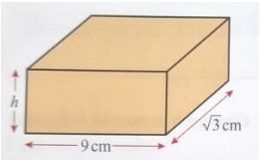
4. There are estimated to be 10^{12} bacteria in a dish. After an antibiotic is added, the number of bacteria is reduced to one millionth of the original. How many bacteria are left?

5. Simplify
   (a) $\frac{12a^{14}b^8}{(3a^4b^3)\times(2a^2b^3)}$
   (b) $\frac{(36m^6n^2)\times(3m^2n^4)}{(3m^2n)^2}$
   (c) $\frac{(4a^2b^3)^2}{\sqrt{(64a^2)}}$
   (d) $\left(3a^{\frac{1}{3}}\right)^3$
   (e) $(3x)^3 \times (2x)^3$
   (f) $(4x)^{-1} \div (8x^2)^{-1}$

6. Eva says that $\sqrt{45}$ is greater than $\sqrt[3]{290}$. Liam does not agree. Explain clearly who is correct, without using a calculator.

7. The volume of this cuboid is 81 cm^3.

   Find the value of h, giving your answer in index notation



8. Find x in the following
   (a) $4^x = 32$
   (b) $25^x = 125$
   (c) $64^x = \frac{1}{4}$
   (d) $2^{4x+1} = 4^{x+1}$
   (e) $9^{2x-1} = 27^{x+2}$
   (f) $32^{3x-1} = 16^{4x-3}$
   (g) $\left(\frac{1}{2}\right)^x = 16$
   (h) $14^x = 1$
   (i) $\left(\frac{1}{2}\right)^x + \left(\frac{1}{5}\right)^x = 29$

9. Sarwan has £(2^{16}) to share equally between 8 people. How much does each person get (leave your answer in index form)?

10. Which is true: $3^{m-1} = \frac{3}{m}$ or $3m^{-1} = \frac{1}{3m}$?

11. Given that $4y^{-2} = 4(y^{-2}) = 4\left(\frac{1}{y^2}\right) = \frac{4}{y^2}$, find the value of:
   (a) $5m^{-2}$ when $m = 2$
   (b) $8y^{-1}$ when $y = 4$
   (c) $10n^{-3}$ when $n = 5$
   (d) $6a^{-2}$ when $a = 4$

12. Evaluate the following:
   (a) $\left(2\frac{1}{4}\right)^{-\frac{1}{2}}$
   (b) $\left(2\frac{7}{9}\right)^{\frac{3}{2}}$
   (c) $\left(\frac{3}{7}\right)^{-2}$
   (d) $\left(2\frac{10}{27}\right)^{-\frac{2}{3}}$

13. (a) Find x if $\frac{8^x \times 64^3}{16^7} = 32$

   (b) Find x if $\frac{81^x \times 9^2}{27^4} = 1$

14. Find x in the following:
    (a) $32^x = 2$
    (b) $81^x = 3$
    (c) $125^x = 5$
    (d) $49^x = 7$
    (e) $121^x = 11$
    (f) $27^x = 3$
    (g) $243^x = 3$
    (h) $256^x = 16$
    (i) $3^x = \frac{1}{3}$
    (j) $81^x = \frac{1}{3}$
    (k) $125^x = \frac{1}{5}$
    (l) $512^x = \frac{1}{2}$

15. Find x the following:
    (a) $m^3 \times \sqrt[3]{m} = m^x$
    (b) $m^x \times \frac{1}{m^4} = \frac{1}{m^5}$
    (c) $\frac{\sqrt{m}}{m^x} = m^2$
    (d) $\sqrt[4]{a} \times (a^x)^2 = a^3$
    (e) $\frac{n^3}{\sqrt{n^x}} = \frac{1}{n^2}$
    (f) $\frac{m^{x-3}}{m^4} = \frac{1}{\sqrt{m}}$

16. Solve
    (a) $4^{\frac{x}{2}} = \frac{1}{32}$
    (b) $49^x = \frac{1}{343}$
    (c) $1000^x = 0.01$
    (d) $2^{x+3} = 4^{2x}$
    (e) $4^x = \left(\sqrt{2}\right)^6$
    (f) $5^{\frac{4x}{3}} = 125^{\frac{2}{3}}$

17. Solve
    (a) $\left(\frac{2}{3}\right)^x = \frac{9}{4}$
    (b) $\left(\frac{3}{5}\right)^x = \frac{27}{125}$
    (c) $\left(\frac{2}{5}\right)^x = \frac{125}{8}$
    (d) $\left(\frac{10}{7}\right)^x = \frac{49}{100}$
    (e) $\left(\frac{9}{4}\right)^x = \frac{32}{243}$
    (f) $\left(\frac{1}{2}\right)^x = 1024$
    (g) $\left(\frac{8}{27}\right)^x = \frac{81}{16}$
    (h) $\left(\frac{3}{4}\right)^x = \frac{16}{9}$
    (i) $\left(\frac{8}{125}\right)^x = \frac{25}{4}$